

Lab Session 04

Practice behavioral view UML diagrams for the suggested system: State Chart, Sequence and Collaboration Diagrams

State chart diagram

State machine diagram is a **behavior diagram** which shows discrete behavior of a part of designed system through finite state transitions. State machine diagrams can also be used to express the usage protocol of part of a system.

The UML defines the Simple State, Composite State and Submachine state.

Simple State

A **simple state** is a state that does not have sub-states - it has no **regions** and it has no **submachine states**. Simple state is shown as a rectangle with rounded corners and the state name inside the rectangle. Simple state may have compartments. The compartments of the state are:

- name compartment
- internal activities compartment
- internal transitions compartment

Name compartment holds the (optional) name of the state, as a string. States without names are called **anonymous states** and are all considered distinct (different) states. Name compartments should not be used if a name tab is used and vice versa.

Internal activities compartment holds a list of internal actions or state (do) activities (behaviors) that are performed while the element is in the state. The activity label identifies the circumstances under which the behavior specified by the activity expression will be invoked. The behavior expression may use any attributes and association ends that are in the scope of the owning entity. For list items where the expression is empty, the slash separator is optional.

Several labels are reserved for special purposes and cannot be used as event names. The following are the reserved activity labels:

- entry (behavior performed upon entry to the state)
- do (ongoing behavior, performed as long as the element is in the state)
- exit (behavior performed upon exit from the state)

Internal transition compartment contains a list of internal transitions, where each item has the form as described for trigger. Each event name may appear more than once per state if the guard conditions are different. The event parameters and the guard conditions are optional. If the event has parameters, they can be used in the expression through the current event variable.

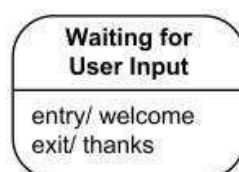


Fig 5.1 Simple state with name and internal activities compartments

Creating State Chart Diagram in StarUML

To create a State-chart Diagram:

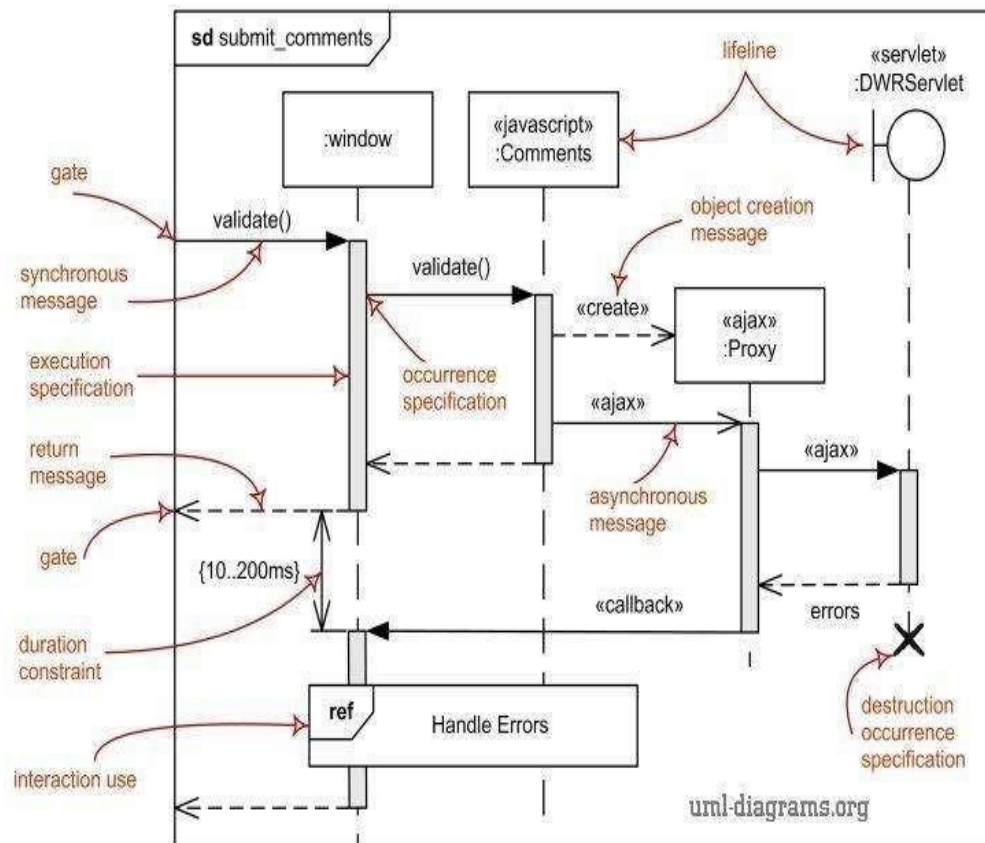
1. Select first an element where a new Statechart Diagram to be contained as a child.
2. **Model | Add Diagram | Statechart Diagram** in Menu Bar or select **Add Diagram | Statechart Diagram** in Context Menu.
3. Toolbox contains all the required options namely simple states, internal activities (entry, do and exit), transitions, initial and final states etc. to create the required state chart diagram.

Sequence diagram

Sequence diagram is the most common kind of interaction diagram, which focuses on the message interchange between a numbers of lifelines.

Sequence diagram describes an interaction by focusing on the sequence of messages that are exchanged, along with their corresponding occurrence specifications on the lifelines.

Major elements of the sequence diagram are shown on the picture below.



Major elements of UML sequence diagram.

Create Sequence Diagram

To create a Sequence Diagram:

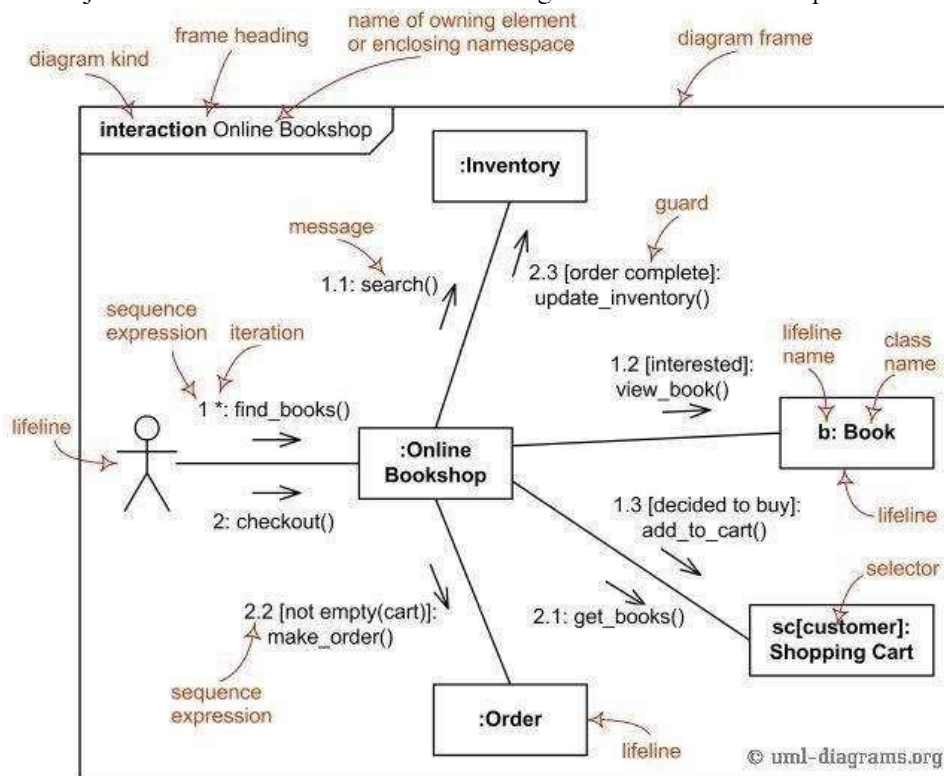
1. Select first an element where a new Sequence Diagram to be contained as a child.
2. Select **Model | Add Diagram | Sequence Diagram** in Menu Bar or select **Add Diagram | Sequence Diagram** in Context Menu.
3. All the major elements of UML sequence diagram (shown above) are available in the toolbox.

Collaboration or communication diagram

Communication diagram (called **collaboration diagram** in UML 1.x) is a kind of UML **interaction diagram** which shows interactions between objects and/or **parts** (represented as **lifelines**) using sequenced messages in a free-form arrangement.

Communication diagram corresponds (i.e. could be converted to/from or replaced by) to a simple **sequence diagram** without structuring mechanisms such as interaction uses and combined fragments. It is also assumed that **message overtaking** (i.e., the order of the receptions are different from the order of sending of a given set of messages) will not take place or is irrelevant.

The following nodes and edges are drawn in a UML communication diagrams: **frame**, **lifeline**, and **message**. These major elements of the communication diagram are shown on the picture below:



The major elements of UML communication diagram.

Create Communication/Collaboration Diagram

To create a Communication Diagram:

1. Select first an element where a new Communication Diagram to be contained as a child.
2. Select **Model | Add Diagram | Communication Diagram** in Menu Bar or select **Add Diagram | Communication Diagram** in Context Menu.
3. All the major elements of UML sequence diagram (shown above) are available in the toolbox.

Bank Automated Teller Machine (ATM) System

State Transition / State Chart Diagram

Purpose: An example of UML behavioral state machine diagram describing Bank Automated Teller Machine (ATM) top level state machine.

Summary: ATM is initially turned off. After the power is turned on, ATM performs startup action and enters **Self Test** state. If the test fails, ATM goes into **Out of Service** state, otherwise there is **triggerless transition** to the **Idle** state. In this state ATM waits for customer interaction.

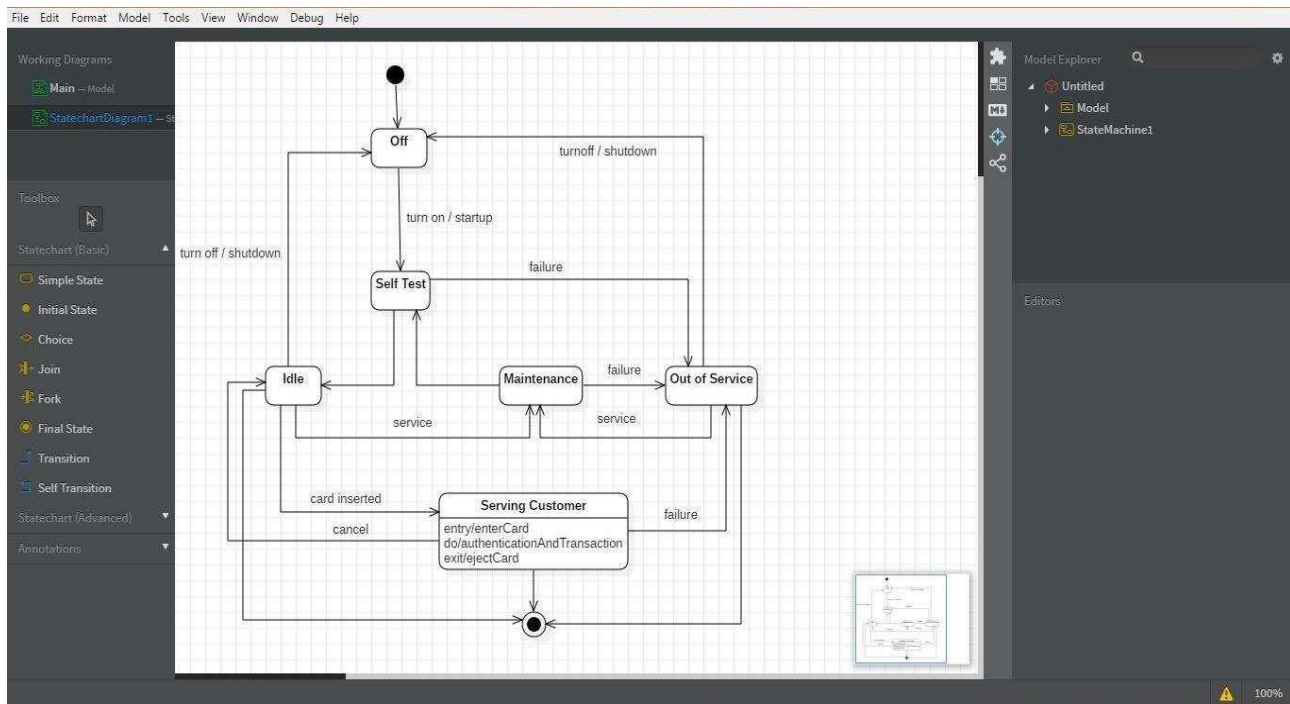


Fig 5.4 State Chart Diagram for ATM System

Sequence and Communication Diagram

Here, customer inserts his card in the ATM machine. The card is verified through bank and after authentication is done, customer is allowed to do transactions. He chooses to withdraw some cash. His request is processed and amount is debited from his account. The customer receives cash, card and receipt from the respective slots.

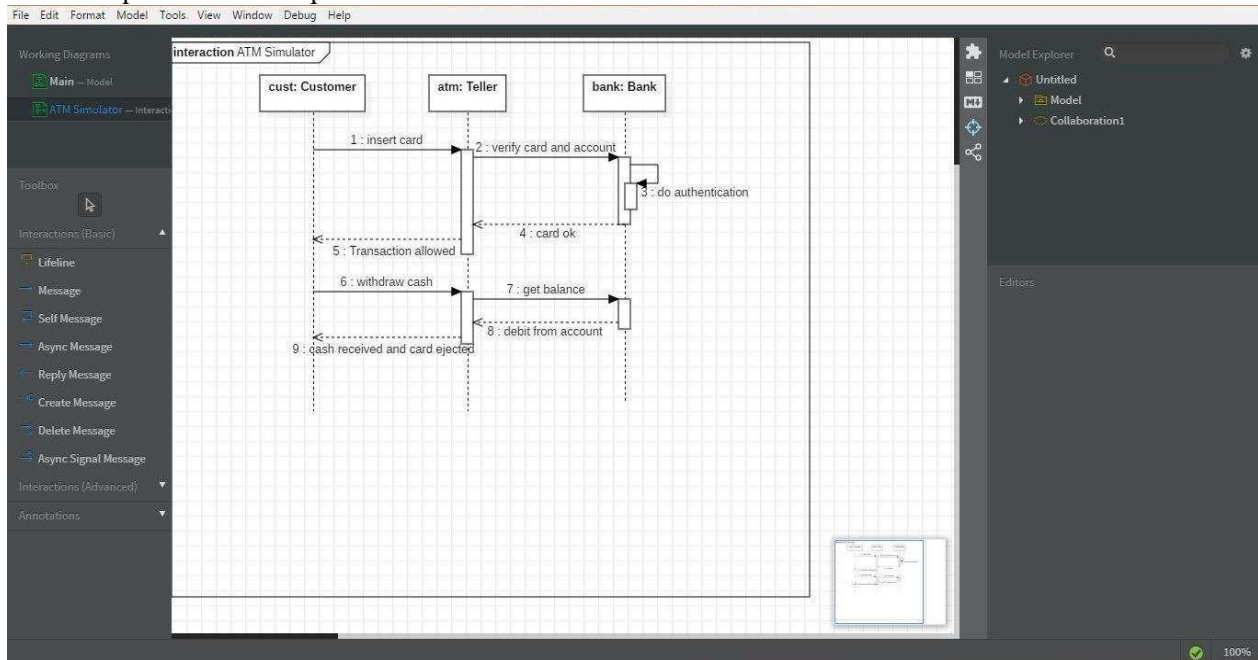


Fig 5.4 UML Sequence Diagram for ATM System

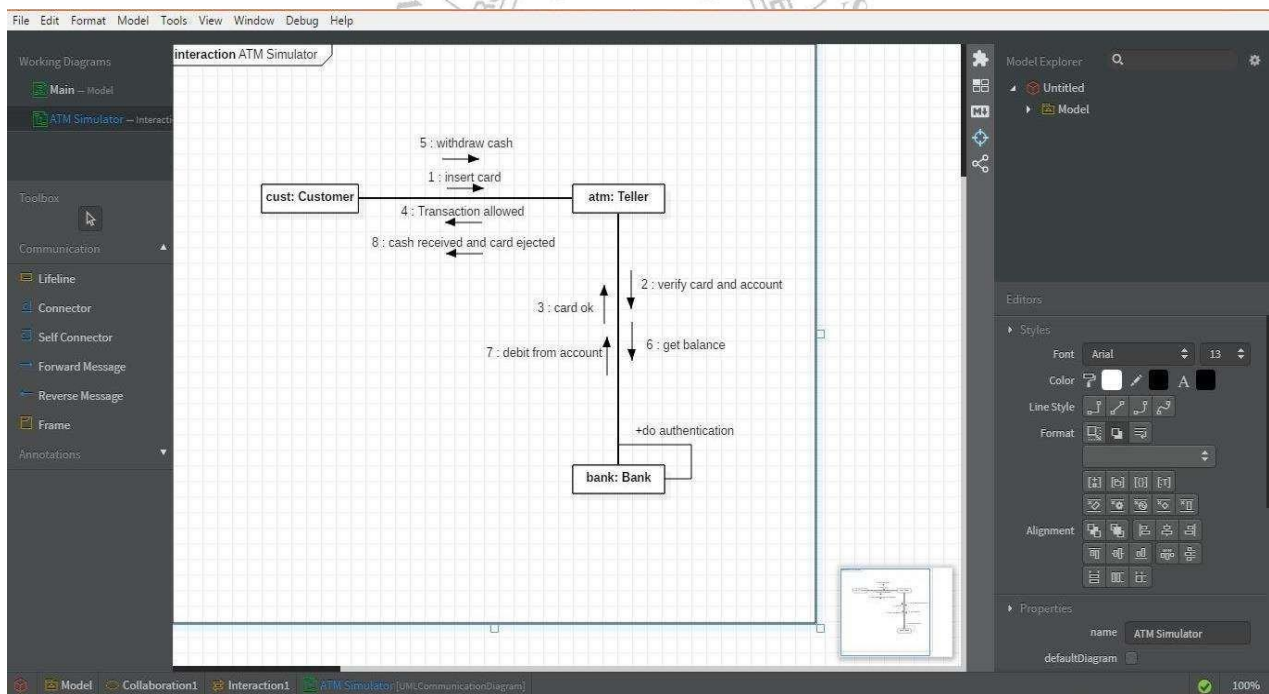


Fig 5.5 UML Collaboration Diagram for ATM System

EXERCISE

1. Create a State Chart Diagram for Microwave Oven Operation using StarUML. Also attach print out.
2. Create Sequence and Collaboration diagrams for Online Bookshop where the Web customer actor can search inventory, view and buy books online. Also attach the printout.
3. UML diagrams can also be created by means of variety of different online tools. Visual Paradigm is one of them. Create any three UML diagrams using this online tool for the ‘Online Bookshop System’. Attach print out